

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T40 · Angles in Polygons and Parallel Lines

Questions

Student book · questions with writing space

■ 16 pages

FAMILY MAP

Angles in Polygons Parallel Lines

F01	Use regular and general polygon angles	11 Q 46 marks
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F02	Use quadrilateral properties	2 Q 9 marks
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F03	Use triangle angle properties	1 Q 4 marks
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F04	Use basic angle and line facts	1 Q 5 marks
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Question 14

4MA1-2H Q3

MODULAR UNIT 1

4 marks

- 3 $ABCD$ is a trapezium.

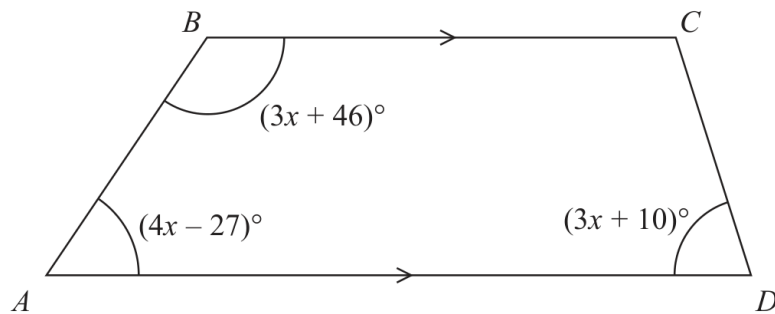


Diagram **NOT**
accurately drawn

BC is parallel to AD

Find the size of the largest angle inside the trapezium.

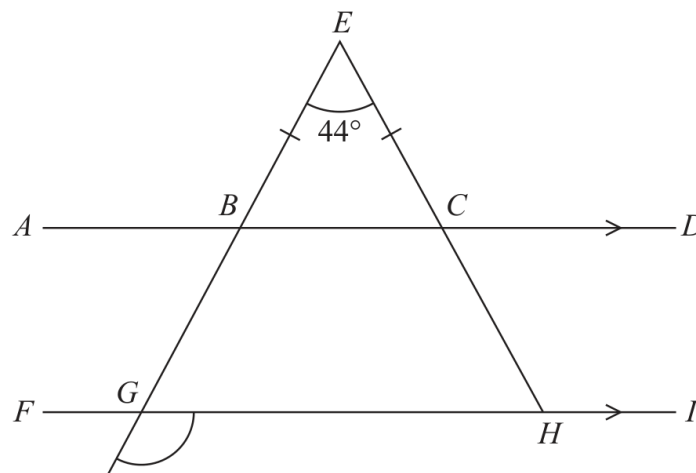
..... °

A triangle is shown with three interior angles. The top angle is labeled 30° . The bottom-left angle is labeled $(4x + 10)^\circ$. The bottom-right angle is labeled $(x + 20)^\circ$.

Diagram **NOT**
accurately drawn

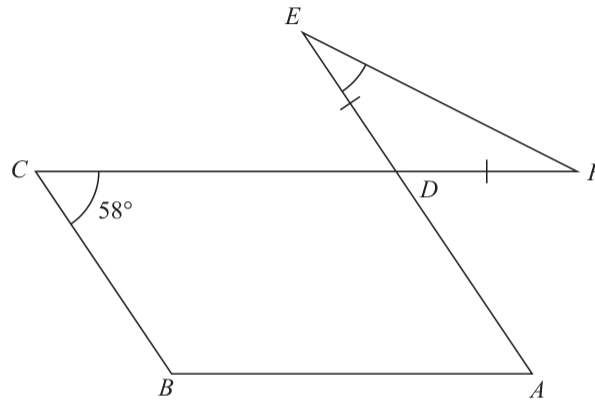
Work out the value of x .

Diagram **NOT**
accurately drawn



Work out the size of angle JGH .
Give a reason for each stage of your working.

Diagram **NOT**
accurately drawn



Give a reason for each stage of your working.

3

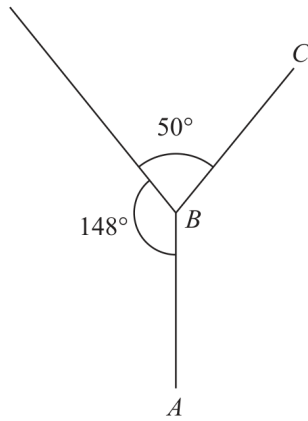


Diagram **NOT**
accurately drawn

AB and BC are two sides of a regular polygon with n sides.

Work out the value of n
Show your working clearly.

$n = \dots\dots\dots$

(Total for Question 3 is 4 marks)

Question 04

4MA1-1H Q5

MODULAR UNIT 2

4 marks

- 5 In the diagram, $ABCDE$ is a regular pentagon.

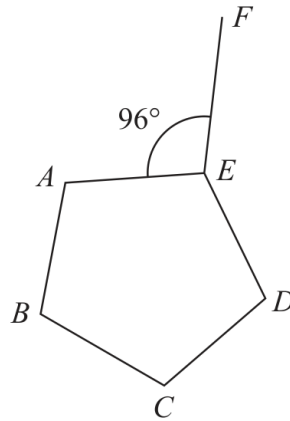


Diagram **NOT**
accurately drawn

Angle $AEF = 96^\circ$

Work out the size of the obtuse angle FED
Show your working clearly.

Question 24

4MA1-2H Q11

MODULAR UNIT 2

3 marks

- 11 Here is a 9-sided regular polygon $ABCDEFGHIJ$, with centre O

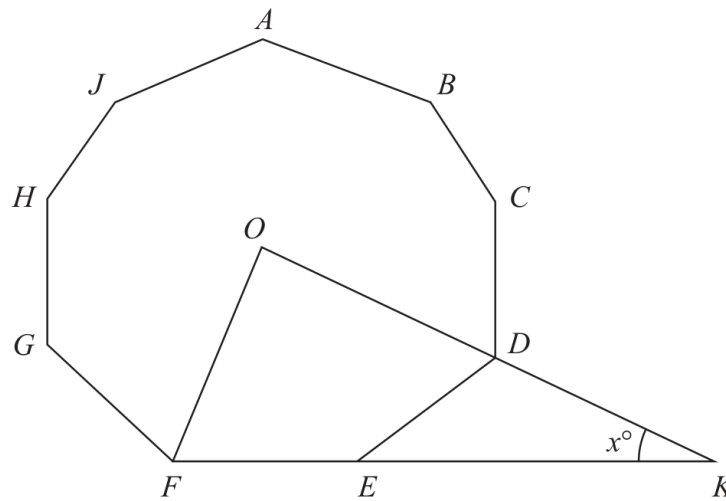


Diagram **NOT**
accurately drawn

ODK and FEK are straight lines.

Work out the value of x

Question 25

4MA1-2H Q11 M01

MODULAR UNIT 2

5 marks

11

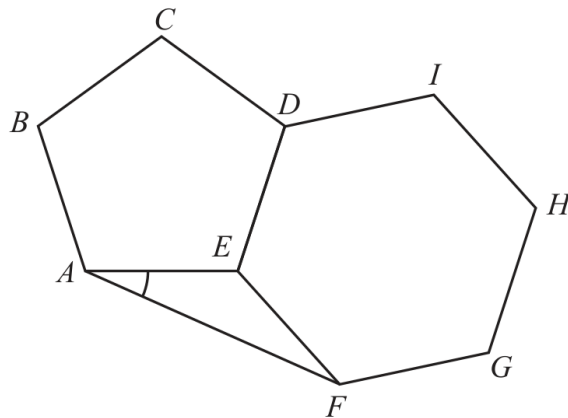


Diagram **NOT**
accurately drawn

$ABCDE$ is a regular pentagon.

$DEFGHI$ is a regular hexagon.

AF is a straight line.

Work out the size of angle EAF

Question 05

4MA1-1H Q4 Q4

MODULAR UNIT 2

4 marks

- 4 The diagram shows parts of three regular polygons, **A**, **B** and **C**, meeting at a point.

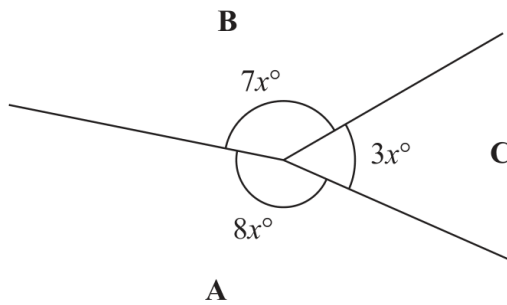


Diagram **NOT**
accurately drawn

Polygon **B** has n sides.

Work out the value of n .

$n = \dots\dots\dots$

Question 19

4MA1-1H Q23 Q23

MODULAR UNIT 2

6 marks

23 A polygon has n sides, where $n > 5$

When arranged in order of size, starting with the largest number, the sizes of the interior angles of the polygon, in degrees, are the terms of an arithmetic sequence.

Here are the first five terms of this sequence.

177 175 173 171 169

Find the value of n

Show clear algebraic working.

Question 23 continued

$n = \dots\dots\dots$

Question 25

4MA1-2H Q11

MODULAR UNIT 2

4 marks

- 11 The diagram shows a regular 10-sided polygon, $ABCDEFGHIJ$

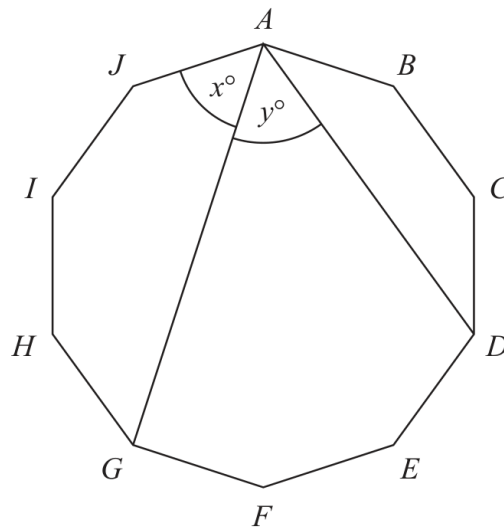
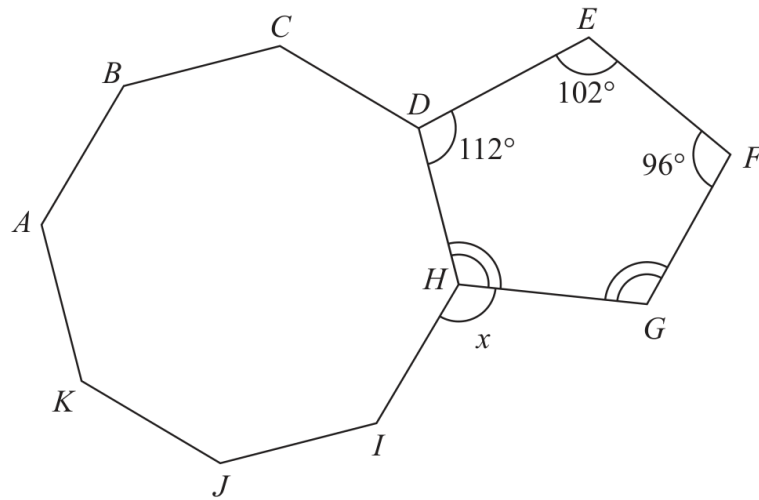


Diagram **NOT**
accurately drawn

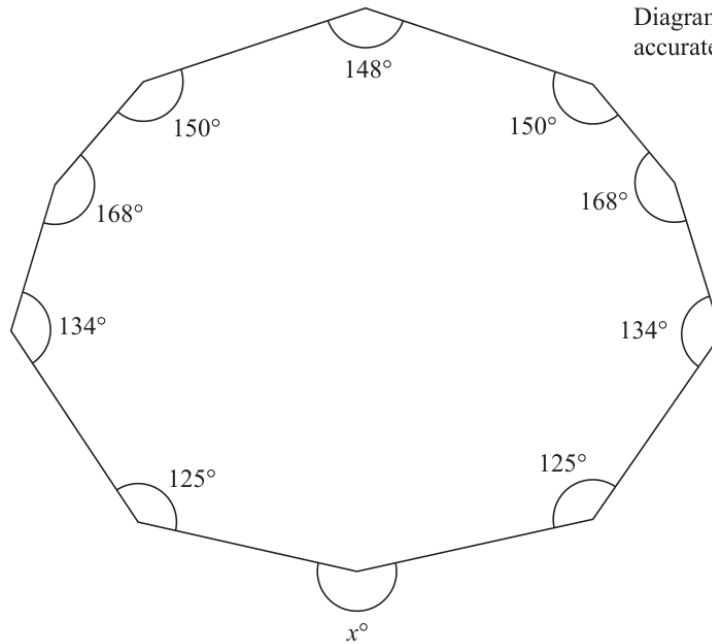
Show that $x = y$

- Diagram **NOT**
accurately drawn



Work out the size of the angle marked x .
Show your working clearly.

Diagram **NOT**
accurately drawn



Work out the value of x .

- 4 The diagram shows a pentagon.

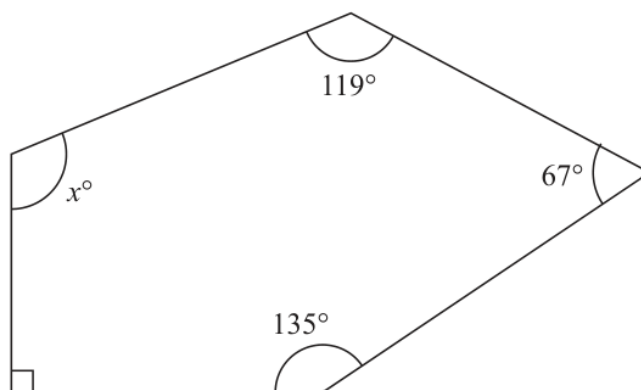


Diagram **NOT**
accurately drawn

Work out the value of x

$x =$

(Total for Question 4 is 3 marks)

The diagram shows part of a regular n -sided polygon with

an interior angle of $(9x + 27)^\circ$

an exterior angle of $(2x - 23)^\circ$

Work out the value of n